

CDFs: CUMULATIVE DISTRIBUTION FUNCTION

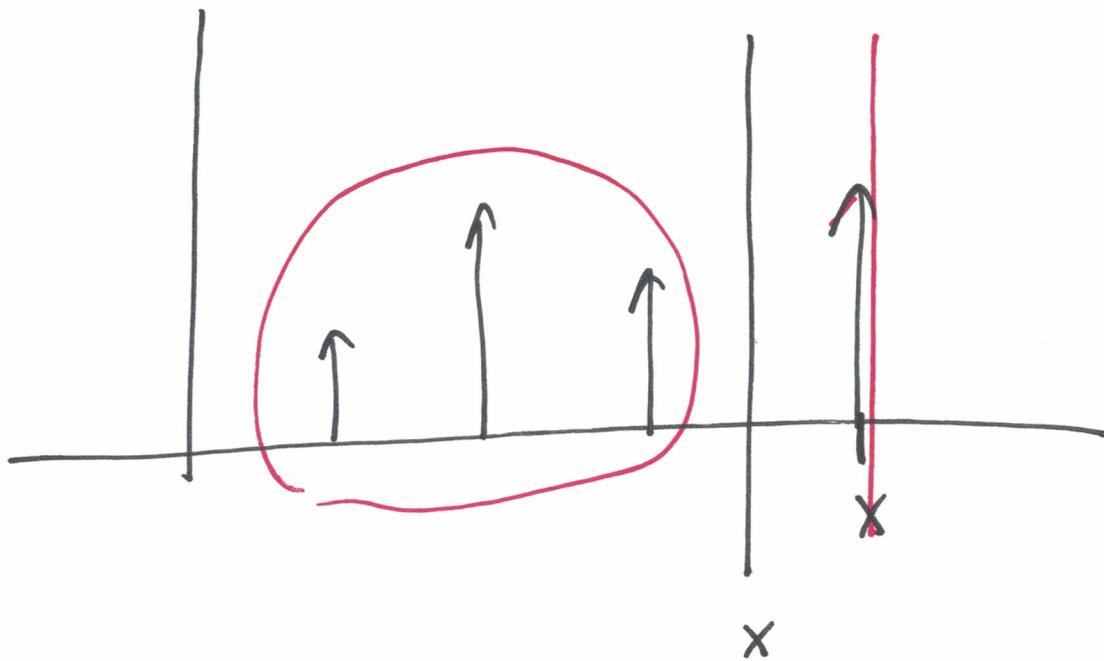
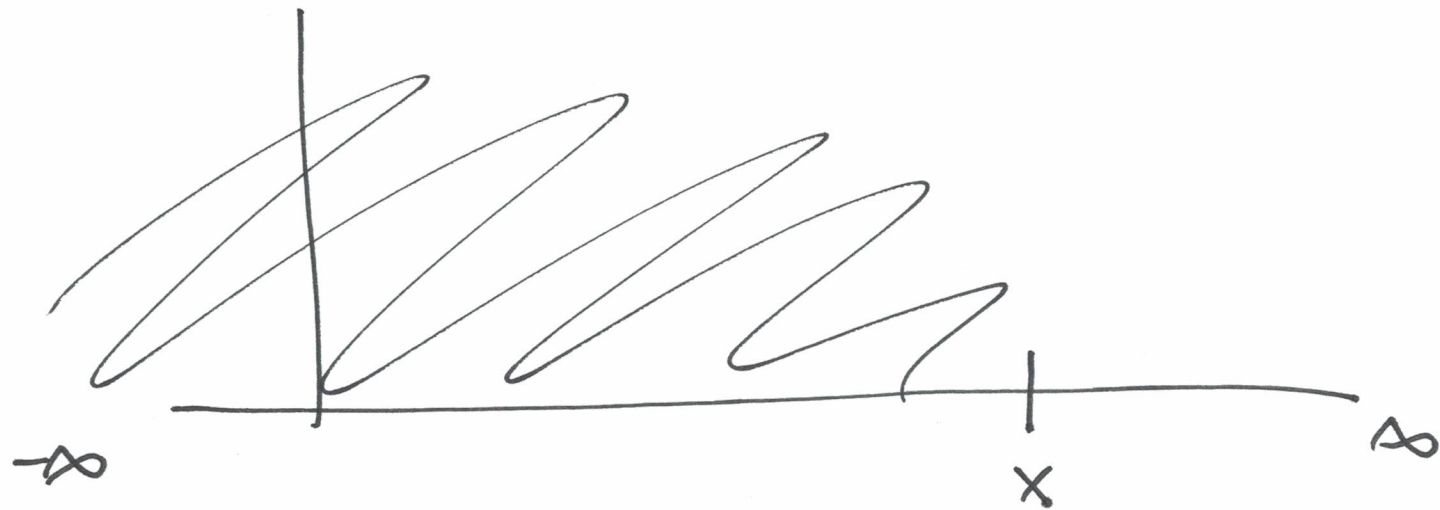
PMF: $P(X=k) = P_X(k)$

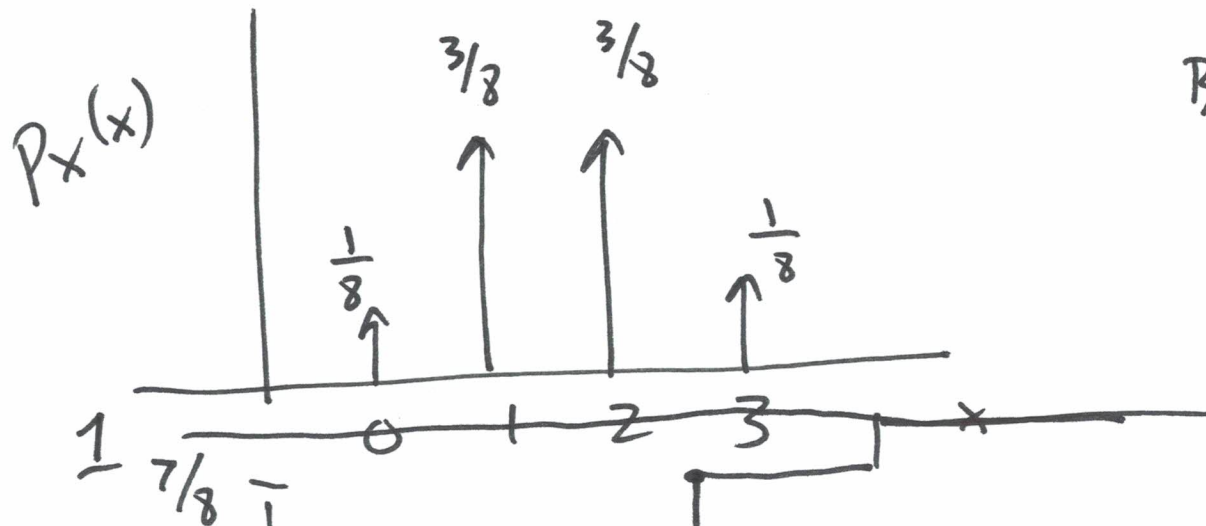
CDF:

$$P(X \leq x) = F_X(x)$$

Diagram illustrating the CDF equation $P(X \leq x) = F_X(x)$ with annotations:

- Red arrow pointing to the x in $P(X \leq x)$: BIG x
- Red arrow pointing to the F in $F_X(x)$: BIG F
- Red arrow pointing to the x in $F_X(x)$: BIG x (NAME OF RANDOM VARIABLE)
- Red arrow pointing to the x in $P(X \leq x)$: LITTLE x (SOME NUMBER)

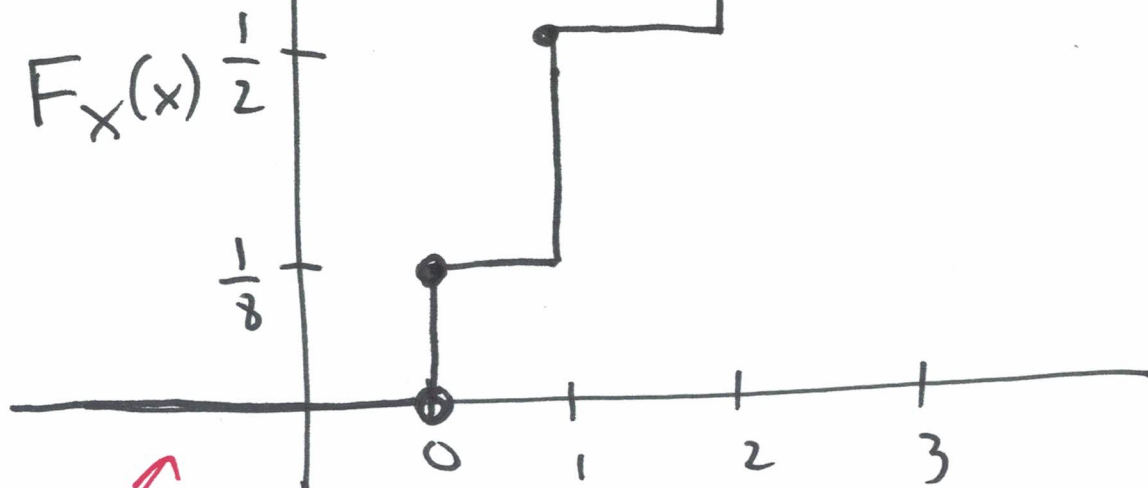




BINOMIAL

$$N=3$$

$$p = \frac{1}{2}$$



CDF AT $+\infty = 1$

CDF

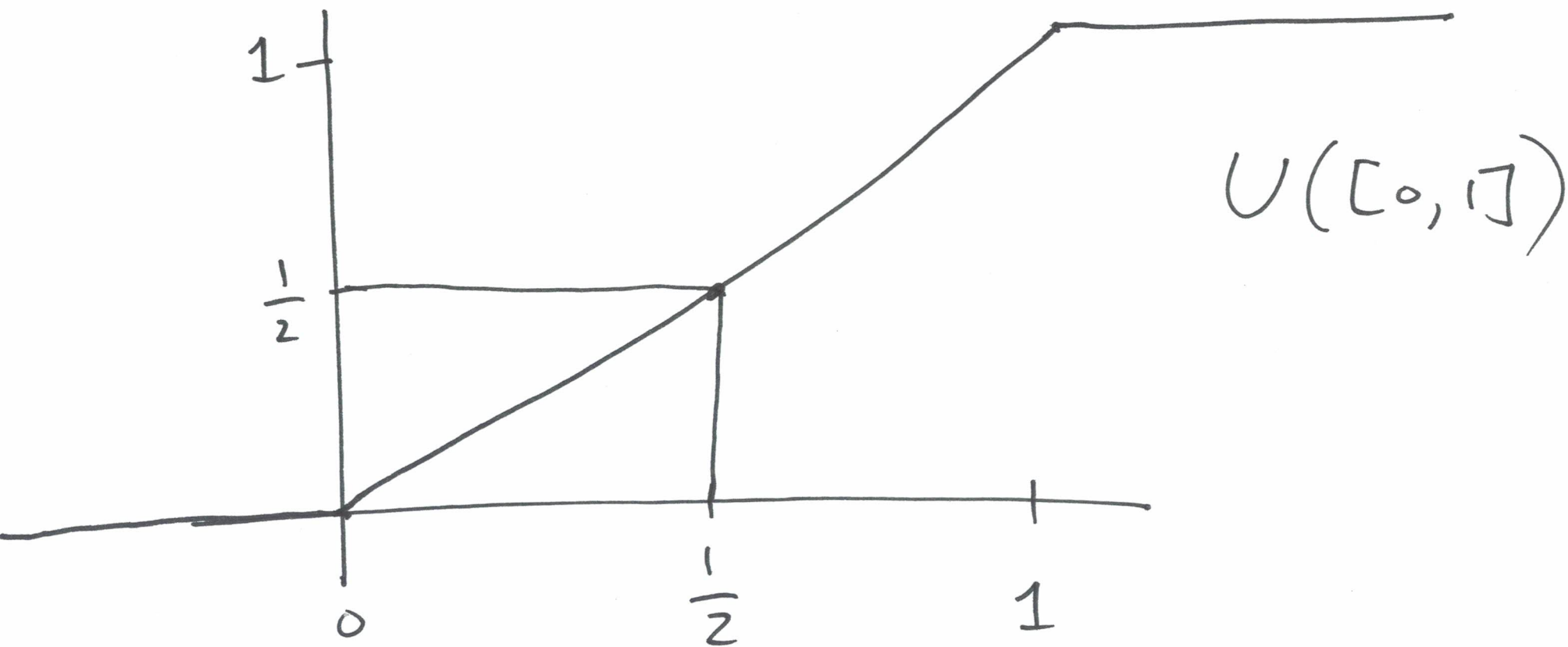
CDF AT $-\infty = 0$

x

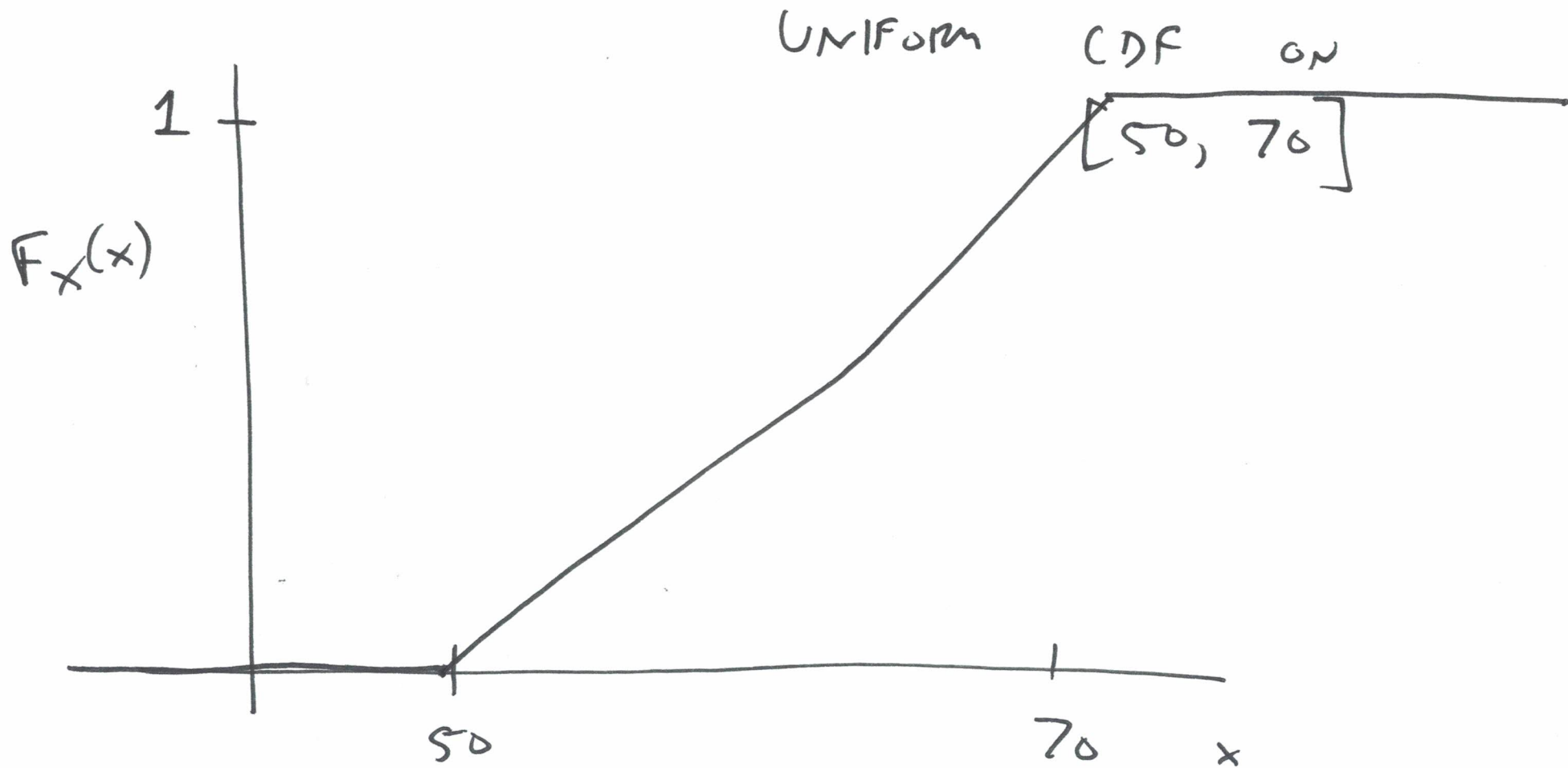
A PIECEWISE
CONSTANT
(STAIRSTEP)

SUM OF STEP FUNCTIONS.

UNIFORM DISTRIBUTION CDF
(CONTINUOUS VERSION)



$$P(X \leq \frac{1}{2}) = F_X(\frac{1}{2}) = \frac{1}{2}$$



EQUATION OF CDF?

$$F_X(x) = \begin{cases} \frac{1}{20}(x-50) & x \in [50, 70] \\ 0 & x < 50 \\ 1 & x > 70 \end{cases}$$

SUPPOSE X HAS CDF:

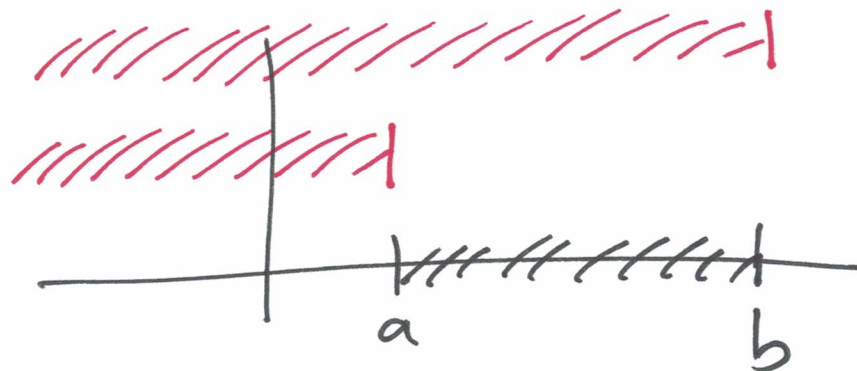
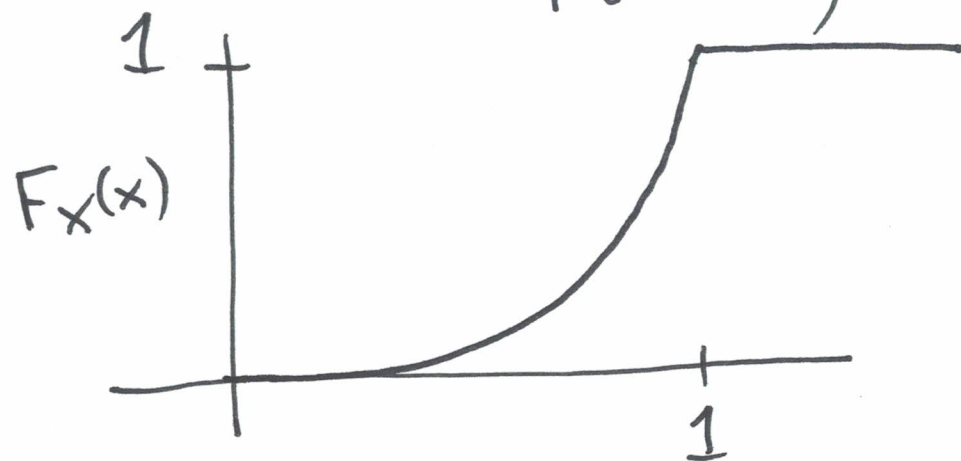
$$F_X(x) = \begin{cases} 0 & x < 0 \\ x^4 & x \in [0, 1] \\ 1 & x > 1 \end{cases}$$

$$P(X \in [\frac{1}{4}, \frac{3}{4}])?$$

$$P(X \in [a, b])$$

$$= P(X \leq b)$$

$$- P(X \leq a) = F_X(b) - F_X(a)$$

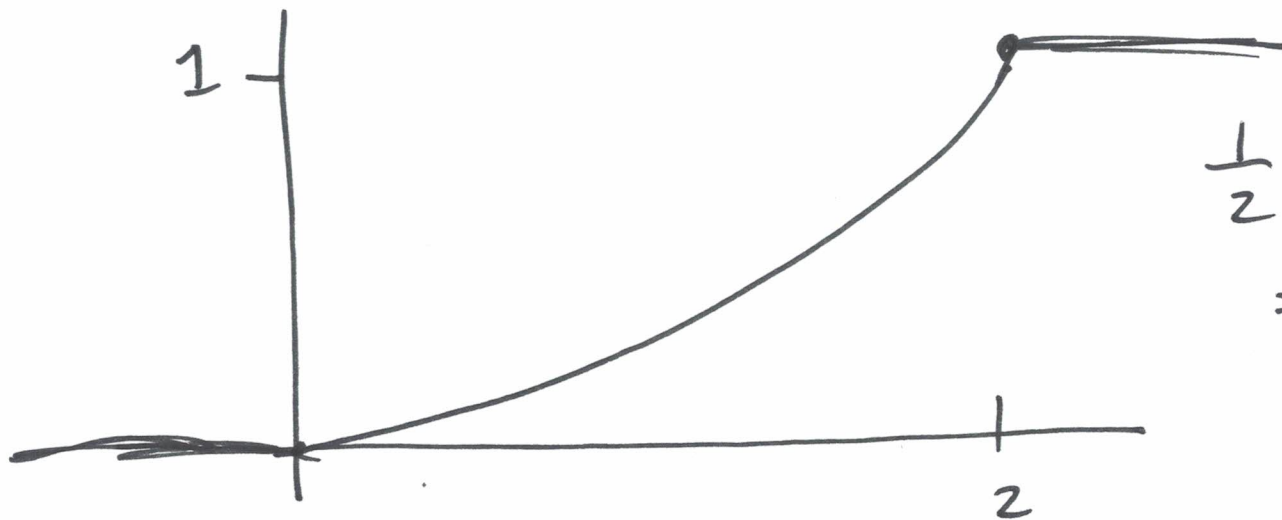


$$\begin{aligned} & F_X\left(\frac{3}{4}\right) - F_X\left(\frac{1}{4}\right) \\ &= \left(\frac{3}{4}\right)^4 - \left(\frac{1}{4}\right)^4 = \frac{80}{256}. \end{aligned}$$

$$P(X > b) = 1 - P(X \leq b) = 1 - F_X(b)$$

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2}cx^2 & x \in [0, 2] \\ 1 & x > 2 \end{cases}$$

WHAT VALUE OF c MAKES THIS A CDF?

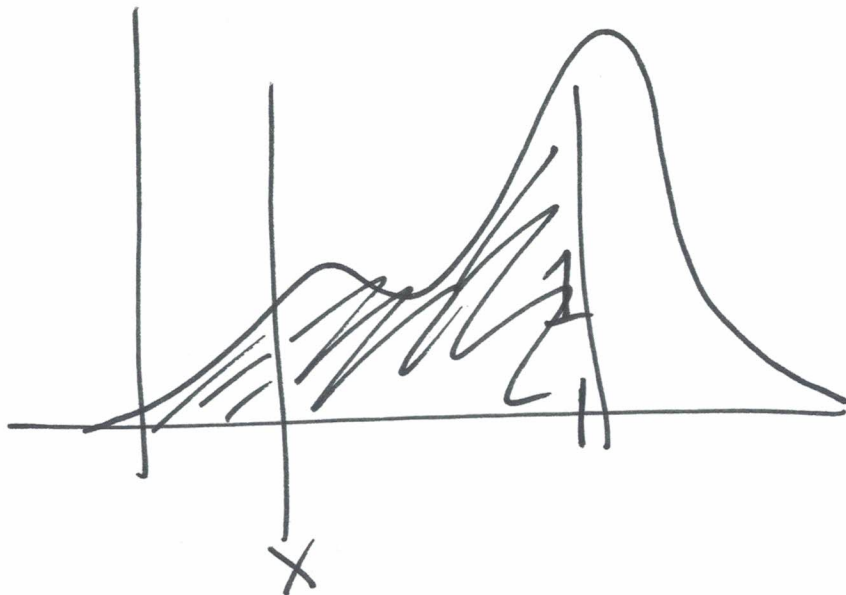


$$\begin{aligned} \frac{1}{2}c(2^2) &= 1 \\ &= 2c = 1 \\ c &= \frac{1}{2}. \end{aligned}$$

$$F_X(x) = \int_{-\infty}^x \underbrace{p_X(x)}_{\text{PMF}} dx$$



THURSDAY

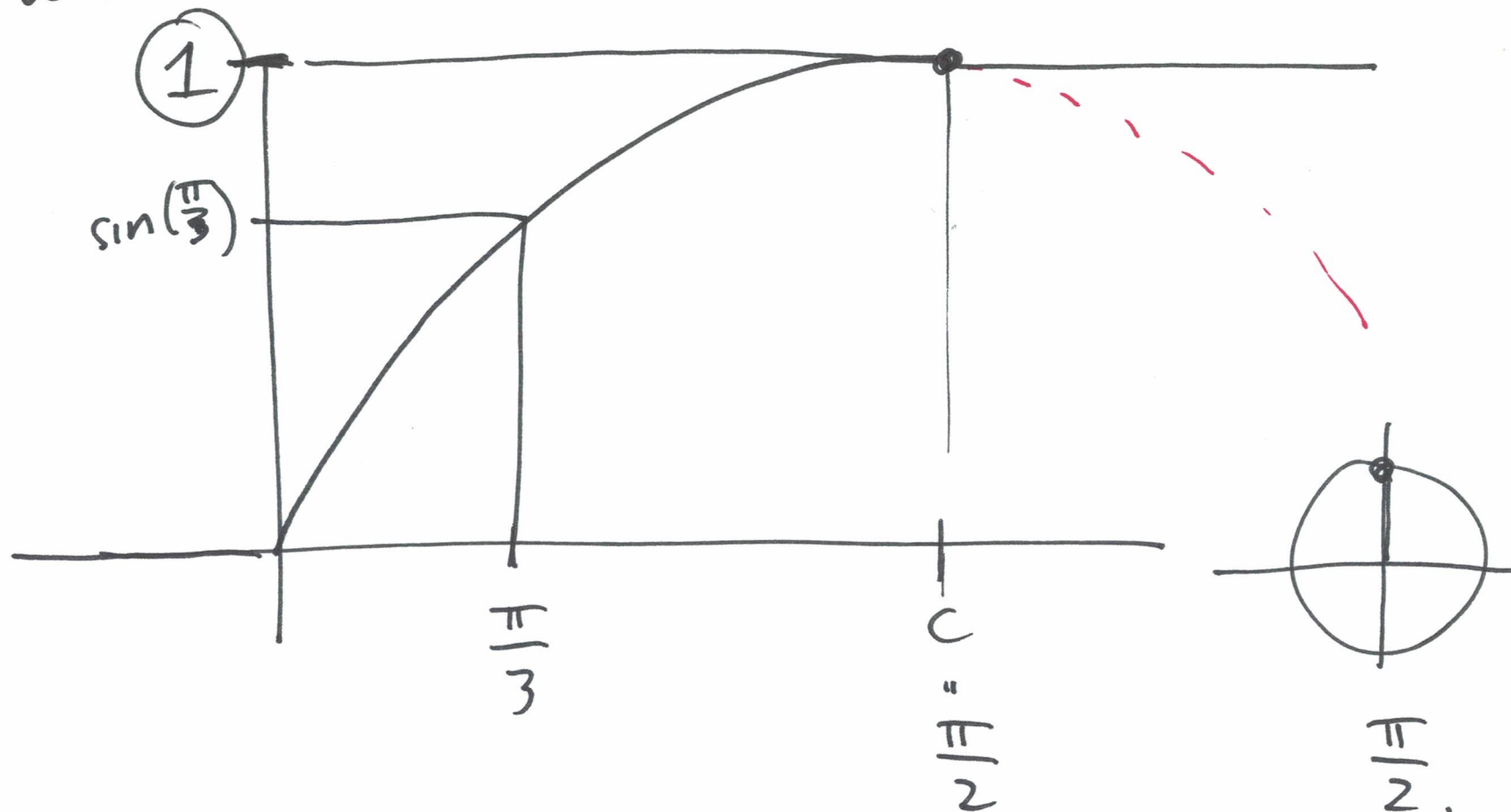


$F_X(x)$
PDF.



$$F_X(x) = \begin{cases} 0 & x < 0 \\ \sin x & x \in [0, c] \\ 1 & x \geq c \end{cases}$$

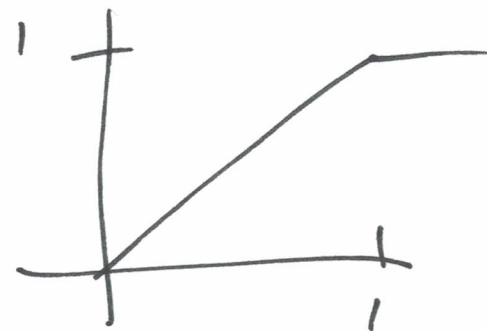
WHAT VALUE OF c ?



X IS A CONTINUOUS UNIFORM RANDOM VARIABLE ON $[0, 1]$.

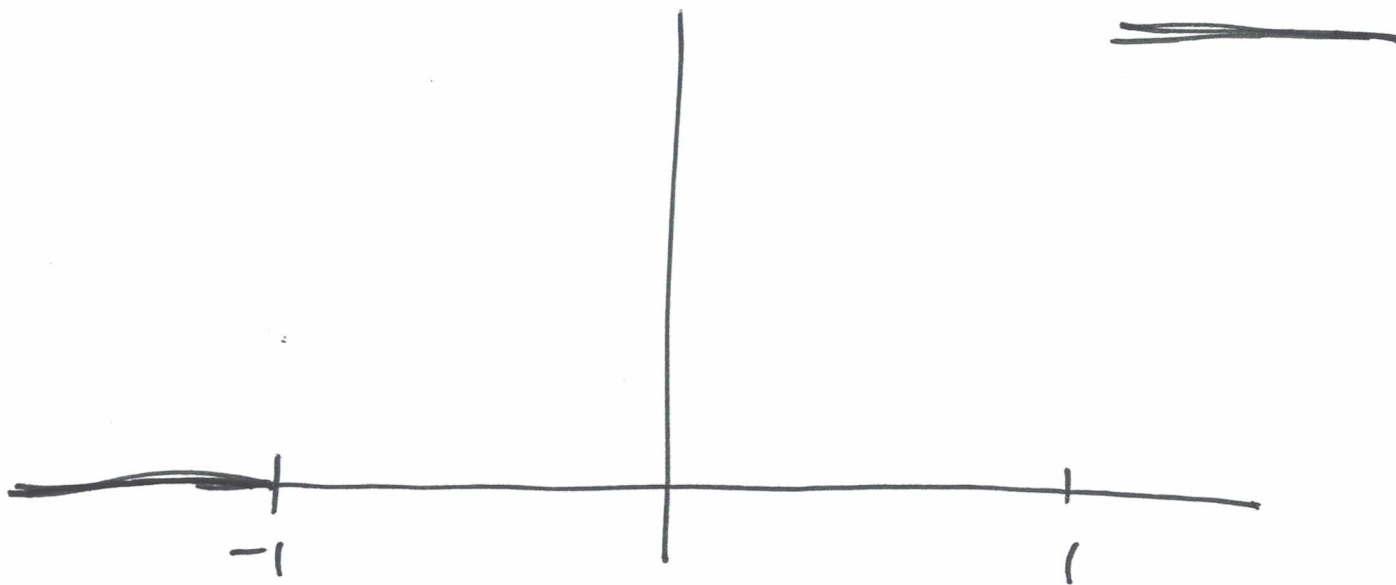
$$Y = 2X^2 - 1.$$

WHAT IS THE CDF OF Y ?



$$X=0 \Rightarrow Y=-1.$$

$$X=1 \Rightarrow Y=+1$$



$$F_Y(y) = P(Y \leq y)$$

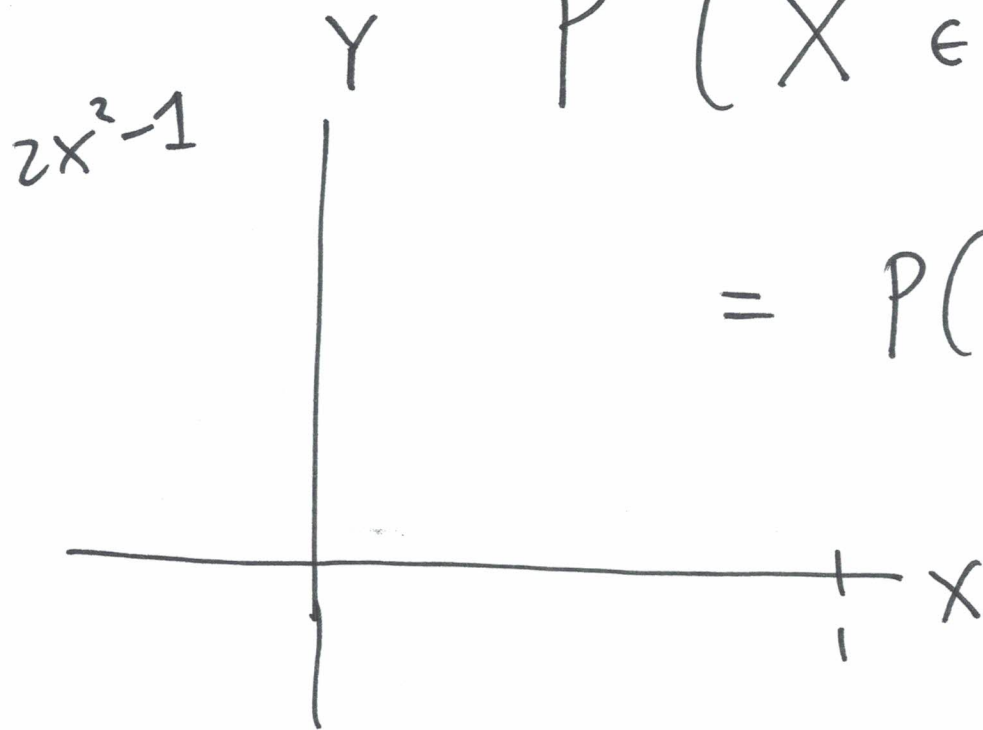
$$P(2X^2 - 1 \leq y)$$

$$P\left(X^2 \leq \frac{y+1}{2}\right)$$

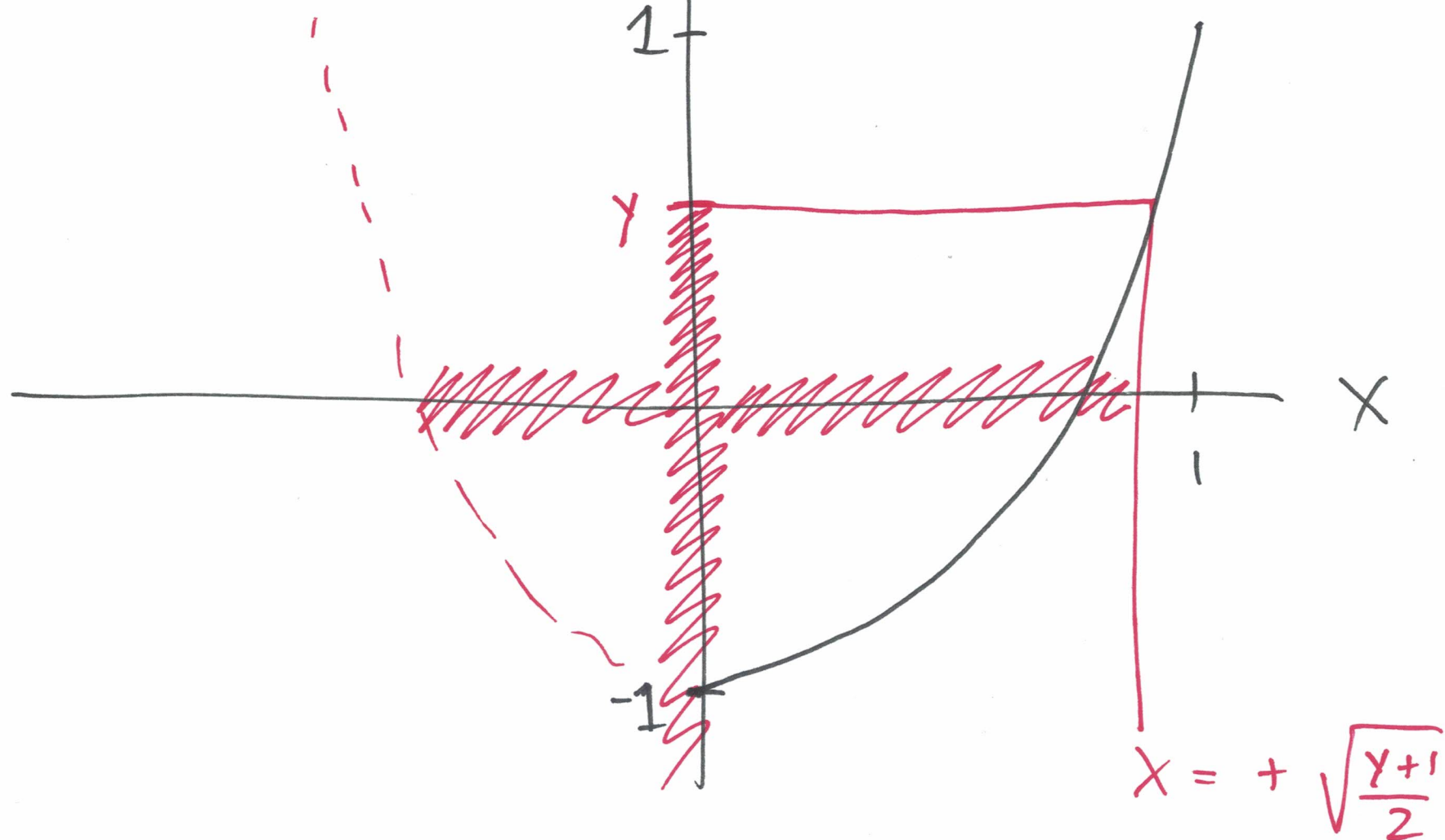
$$P\left(X \in \left[-\sqrt{\frac{y+1}{2}}, +\sqrt{\frac{y+1}{2}}\right]\right)$$

$$= P\left(X \leq \sqrt{\frac{y+1}{2}}\right)$$

$$= \sqrt{\frac{y+1}{2}}$$



$$Y = 2X^2 - 1$$



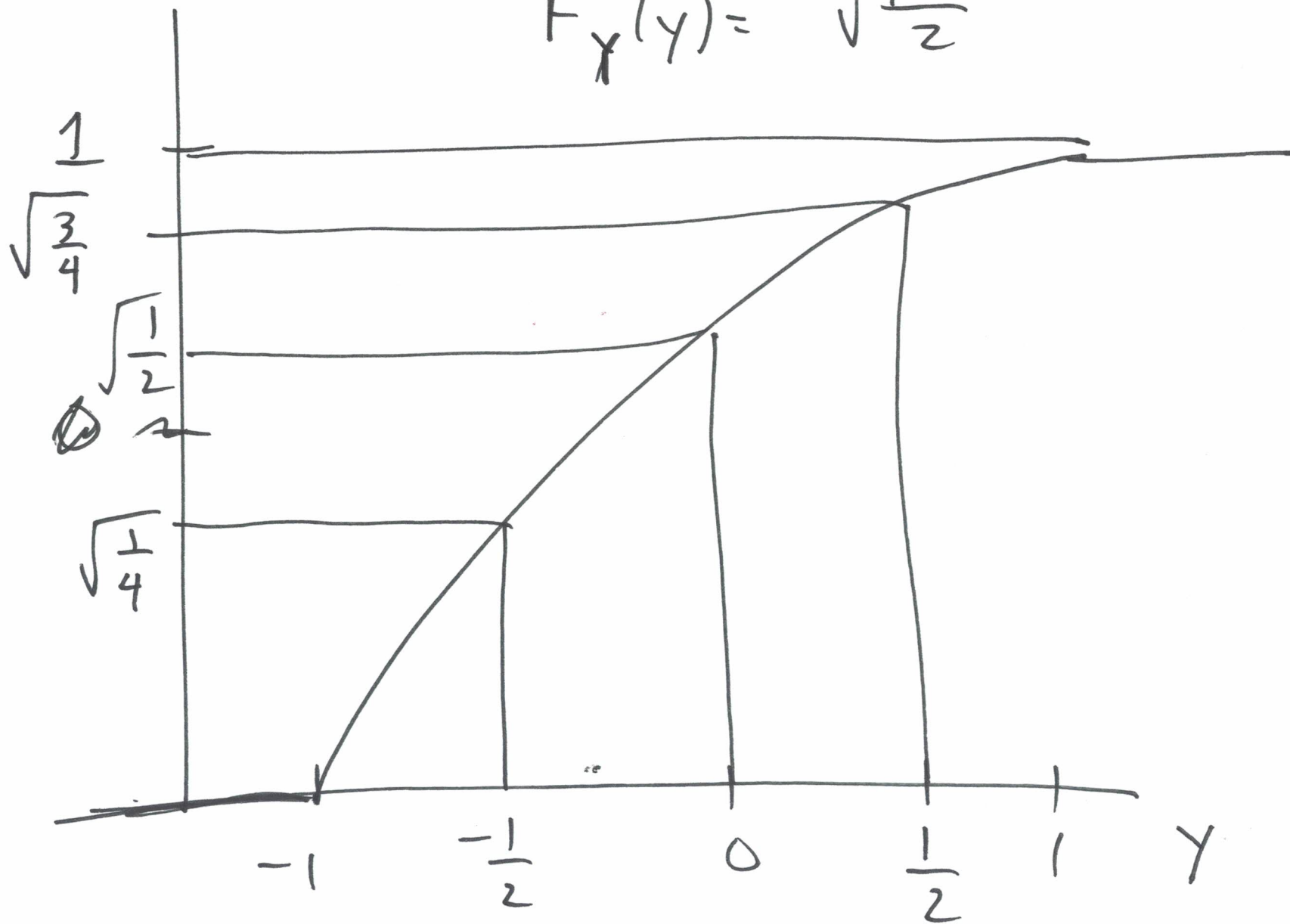
SOME VALUE

$$F_Y(Y) = \begin{cases} 0 & Y < -1 \\ \sqrt{\frac{Y+1}{2}} & Y \in [-1, 1] \\ 1 & Y > 1 \end{cases}$$

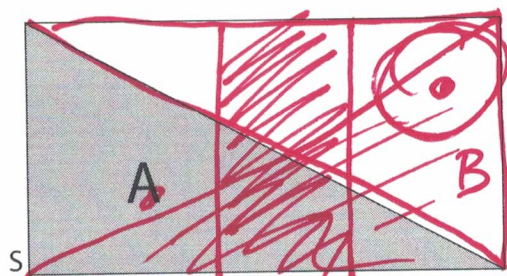
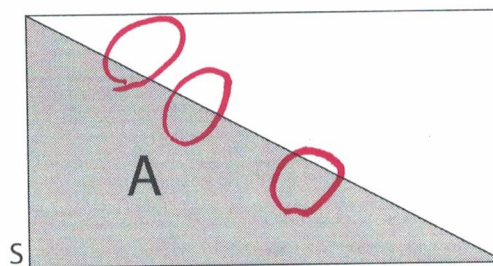
NAME OF R.V.

$$F_Y(a) = \begin{cases} 0 & a < -1 \\ \sqrt{\frac{a+1}{2}} & a \in [-1, 1] \\ 1 & a > 1 \end{cases}$$

$$F_X(y) = \sqrt{\frac{y+1}{2}}$$



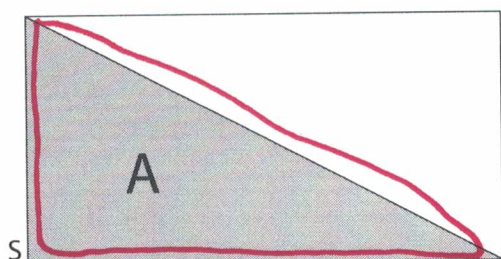
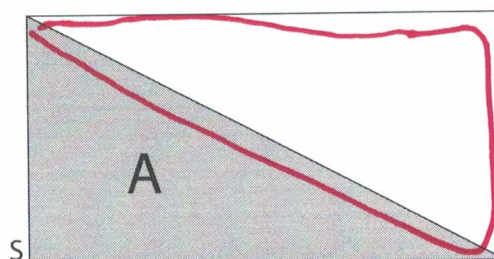
1. (4 points each.) In each picture below, we show the sample space S of an experiment (the outer rectangle) and an event A (the shaded triangle). The distribution of outcomes over the sample space is uniform, so $P(A) = \frac{1}{2}$. In each of the problems below, draw an event B that is **not a subset** of A and satisfies the given requirements. Provide a short explanation for full credit.

(a) A and B are independent.(b) $P(B | A)$ is very small but $P(A | B)$ is very large.

$$P(A \cap B) = P(A)P(B)$$

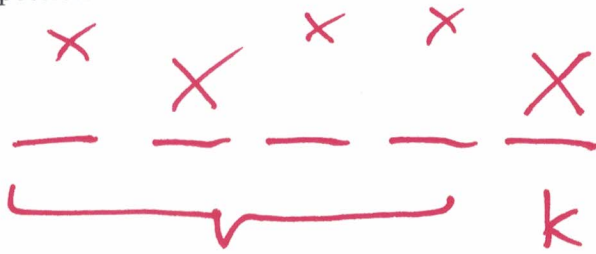
$$\left(\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} \right)$$

→ $\emptyset$

(c) $P(B) \approx P(A)$ and $P(B | A)$ is very large.(d) $P(B) \approx P(A)$ and $P(B | A)$ is very small.

2. (25 points.) Cassian Andor visits a sequence of planets looking for droid friends. Suppose that a droid appears on each planet independently with probability $\frac{1}{4}$. Let X be the random variable corresponding to the planet where Andor encounters the **second droid**.

- (a) (10 points.) Determine a closed-form formula for the probability mass function of X . Be as precise as possible.



$k-1$
EXACTLY 1
droid



$$\binom{n}{2} \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^{k-2}$$

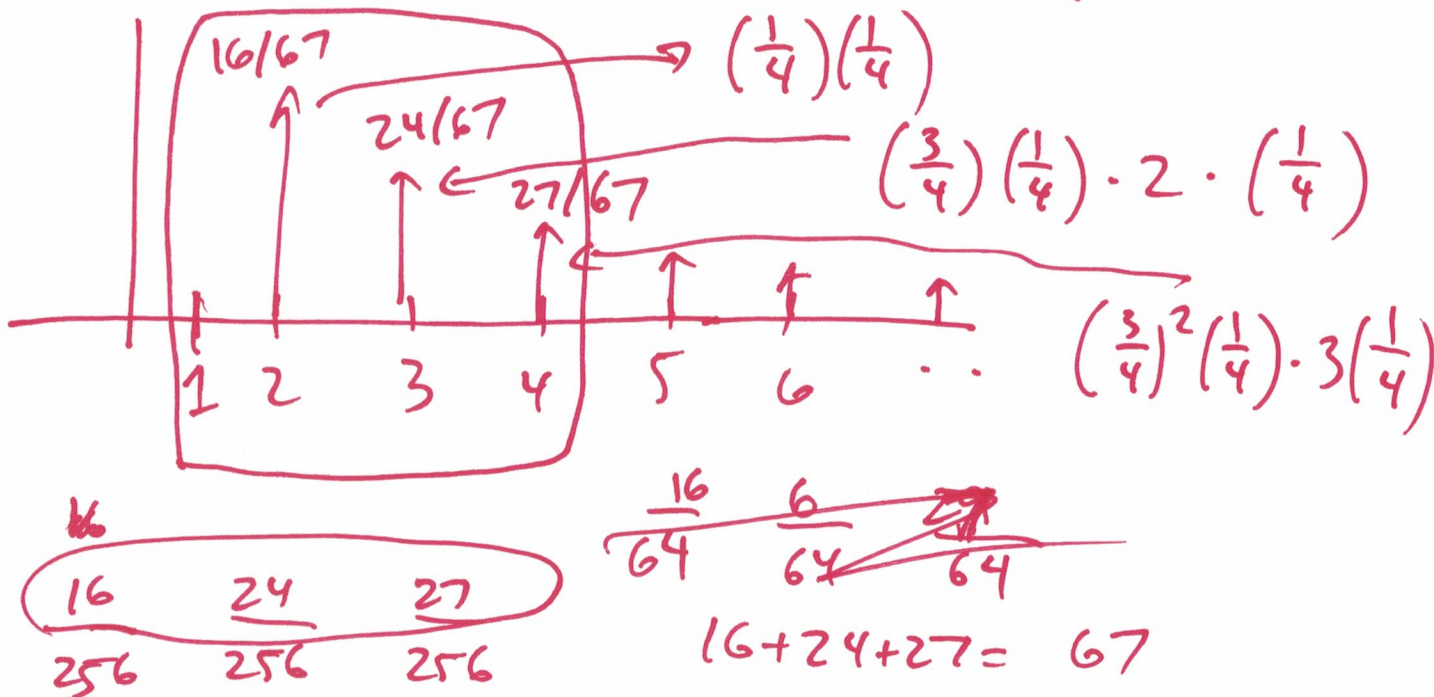
$$P(X=k) =$$

$$(k-1) \left(\frac{1}{4}\right) \left(\frac{3}{4}\right)^{k-2} \left(\frac{1}{4}\right)$$

- (b) (5 points.) Compute the decimal value of $P(X=6)$.

$$5 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^4 = \dots$$

(c) (5 points.) Compute all values for, and produce an accurate sketch of, $p_X(x | X \leq 4)$.



(d) (5 points.) Compute the decimal value of $E(X | X \leq 4)$.

$$2\left(\frac{16}{67}\right) + 3\left(\frac{24}{67}\right) + 4\left(\frac{27}{67}\right) = \dots$$

3. (25 points.) Suppose we are playing the video game Hades, in which we go through a series of rooms. In each room, a single Greek god appears to give us a gift. Suppose we are required to take gifts from **exactly four** gods. There are 12 gods, and each subset of 4 gods you can get gifts from is equally likely.

(a) (5 points.) What is the probability that **both** Artemis and Poseidon are in your chosen set of 4 gods?

$$\binom{12}{4} = \frac{12 \cdot 11 \cdot 10 \cdot 9}{4 \cdot 3 \cdot 2 \cdot 1} = 495$$

A P 

$$\binom{10}{2} = 45$$

$$\frac{45}{495} = \frac{1}{11}$$

(b) (5 points.) What is the probability that **exactly one** of Artemis or Poseidon is in your chosen set of 4 gods?

$$\binom{10}{3} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$$

A — — —
P — — —

$$P(A) + P(P) - 2P(A \cap P)$$

$$\frac{240}{495}$$

(c) (5 points.) Suppose that Artemis is in your set of 4. What is the probability that Poseidon is also in your set?

$$\frac{P(A \cap P)}{P(A)} = \frac{\frac{1}{11}}{\frac{1}{3}} = \frac{3}{11}$$

A — — —
P — — —

- (d) (10 points.) Now suppose our set of 4 gods is fixed to be {Artemis, Hera, Poseidon, Zeus}. We allocate 10 offerings to these gods, giving at least 1 offering to each god so nobody feels left out. For example, one allocation could be $(A, H, P, Z) = (3, 2, 3, 2)$. Suppose each of these possible allocations is equally likely. What is the probability that Hera gets more boons than Zeus?

Hint 1: the number of possible allocations of n offerings to k gods is $\binom{n-1}{k-1}$.

Hint 2: $P(H > Z)$ would need to be the same as $P(Z > H)$ by symmetry.

$$\binom{n-1}{k-1}$$

$$\textcircled{10} \quad 84 - \binom{9}{3} = \frac{3 \cdot 8 \cdot 7}{3 \cdot 2}$$

— — — —
H Z A B

7 1 1 1

①

6 2 1 1
1 1 2
1 2 1

③

5 3 1 1 1
2 2 2
1 3 3
2 2

⑥

$H > Z$	$H = Z$	$Z > H$
	6	

$$\frac{1}{2}(1 - 6)$$

$$\frac{1}{2}\left(1 - \frac{16}{84}\right)$$

4	4	1	1
3	3	3	1
2	2	2	2
1	1	1	3
5	4	3	2
2	3	2	1
2	2	1	8
5	6		

4. (18 points.) Prof. Radke tries to sell his photographs at a local art show. The (continuous) dollar amount it costs him to manufacture a small photograph is a uniform random variable on $[10, 30]$ and the (continuous) dollar amount it costs him to manufacture a medium photograph is a uniform random variable on $[30, 60]$. Prof. Radke charges three times his manufacturing cost plus \$10 for each photograph.

Potential customers arrive at his booth according to a Poisson distribution with an average of one customer every 10 minutes. Each customer has probability 0.2 of buying a photograph. If they buy a photograph, they have probability 0.7 of buying a small photograph and 0.3 of buying a medium photograph.

- (a) (10 points.) What is Prof. Radke's expected **profit** over the course of a 6-hour art show?

$$\begin{array}{ll}
 S: \$20 \text{ TO MAKE} & E(PN=FIT) = 25 + 10 \\
 M: \$45 \text{ TO MAKE} & \quad \quad \quad \$50 S \\
 & \quad \quad \quad \$100 M.
 \end{array}$$

$$E(\# \text{ CUST-MENS}) = 36$$

$$SO \cdot E(S) = 36 \cdot 0.2 \cdot 0.7 = 36 \cdot 1.4 =$$

$$100 \quad E(M) = 36 \cdot 0.2 \cdot 0.3 = 36 \cdot 0.6 =$$

- (b) (8 points.) What is the probability that Prof. Radke sells no photographs during a 2-hour event? Be as precise as you can. Hint: remember the Taylor series $\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$.

$$\alpha = 12$$

$$\frac{12^n}{n!} e^{-12}$$

$$\begin{aligned}
 & P(N=0) + P(\text{NO SALE} | N=1) P(N=1) \\
 & + P(\text{NO SALE} | N=2) P(N=2) \\
 & \quad \quad \quad \underline{\hspace{10em}} \\
 & \quad \quad \quad (0.8)^N
 \end{aligned}$$

5. (16 points.) Jan Dolski collects two types of resources on the surface of a hostile planet: Metals and Organics. Suppose the (integer) number of Metals M_i he collects on day i is a discrete uniform random variable on the range $\{2, 3, 4\}$ and the number of Organics O_i he collects on day i is a binomial random variable with $N = 20$ and $p = \frac{4}{5}$. All of the $\{M_i\}$ and $\{O_i\}$ are independent. Let $M = M_1 + M_2 + M_3 + M_4 + M_5$ and $O = O_1 + O_2 + O_3 + O_4 + O_5$ (that is, the total amount of each resource Jan collects after 5 days).

- (a) (6 points.) Determine $E(O)$ and $\text{Var}(O)$ using properties of expected value and variance.

$$E(O_1 + \dots + O_5) = 5 E(O_i) = 5 \cdot 20 \cdot \frac{4}{5} = 80.$$

$$\begin{aligned} \text{Var}(\dots) &= 5 \text{Var}(O_i) = 5 \cdot 20 \cdot \frac{4}{5} \cdot \frac{1}{5} \\ &= 16. \end{aligned}$$

- (b) (10 points.) Suppose Jan needs 20 Metals to build a battery for his suit. Let $S = M - 20$ be the surplus number of Metals he would have at the end of day 5 (note that S could be negative). Determine $E(S)$ and $\text{Var}(S)$ using properties of expected value and variance.

$$M_1 + \dots + M_5 - 20.$$

$$\begin{aligned} E(M) &= 5 E(M_i) - 20. \\ &= 5(3) - 20 = -5 \end{aligned}$$

$$\begin{aligned} \text{Var}(M) &= 5 \text{Var}(M_i) = 5 \cdot \frac{4^2 - 1}{12} \\ &= 5 \cdot \frac{3}{12}. \end{aligned}$$

$$\binom{n-1}{1} \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^{n-2} \cdot \frac{1}{4}$$

$$* \left| \begin{array}{cccc} * & * & * & * \end{array} \right| * \left| \begin{array}{ccc} * & * & * \end{array} \right| * \dots$$

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{12^n}{n!} e^{-12} \cdot (0.8)^n &= e^{-12} \sum_{n=0}^{\infty} \frac{(9.6)^n}{n!} \\ &= e^{-12} \cdot e^{9.6} = \dots \end{aligned}$$

$$E(5M) = 5E(M)$$

$$\text{Var}(\overbrace{5M}^{\uparrow}) = 25 \text{Var}(M)$$

$$M_1 + M_2 + M_3 + M_4 + M_5$$

$$\text{Var}(M_1) + \text{Var}(M_2) + \dots + \text{Var}(M_5)$$

$$M(k) = \begin{cases} E_k \\ 2210 \end{cases}$$

$$p_X(k) = \begin{cases} \text{---} & k \in 2, 3, \dots \\ 0 & \text{OTHER} \end{cases}$$

Exam 1 scores

